“Let’s not ask our brains to be in two places at once. During lectures and class discussions, close your laptop and be fully present – we will open it again when it is time to code.”

Aalto-yliopisto
Insinööritieteiden
korkeakoulu

L0: Course Overview

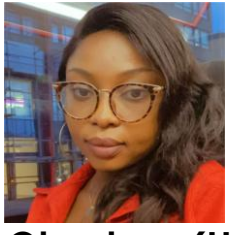
MEC-E2016 Marine Data Analysis
Date: 02.09.2026

Responsible teacher information

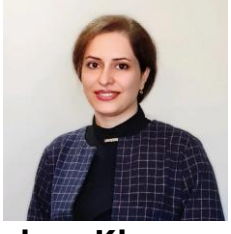
Mashrura Musharraf
Assistant Professor
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Walk-ins are perfectly fine if I am available, but agreeing on a time over email works better

TA information



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Learning Objectives of the Course

Aim: To be able to develop and communicate a complete marine data analysis workflow.

After completing the course, students should be able to –

- Pre-process and visualize marine data
- Apply different classification, regression, and clustering algorithms in marine data analysis
- Evaluate different data analysis techniques and select appropriate technique(s) for chosen engineering problems
- Program the different data analysis algorithms in Python

Cumulative Learning

We go at a pace that makes sense and adjust the depth/breadth of topics

Period I

- L1: Course orientation and foundations of marine data analysis
- L2: Exploratory Data Analysis (EDA)
- L3 & L4: Data pre-processing
- L5: From engineering questions to analytical models
- L6: Marine data analysis in practice

Period II

- L7: Regression modelling, evaluation, and refinement
- L8 & L9: AIS data analytics
- L10 & L11: Cluster analysis
- L12: AI-assisted qualitative data analysis
- Project demonstration

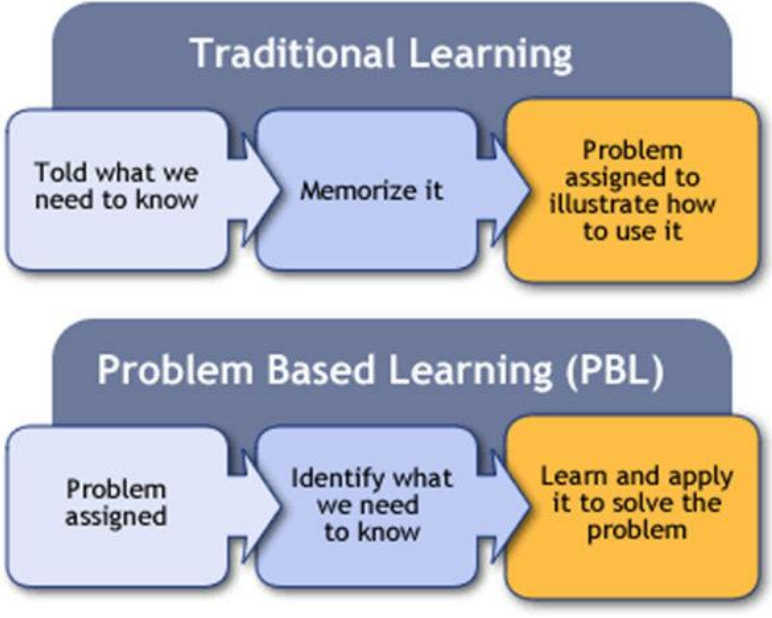
Important Related Concepts

- Fundamentals of statistics
- Problem solving through programming
- Data analysis

We aim to communicate complex ideas in an easy manner and create an **inclusive** learning experience for everyone.

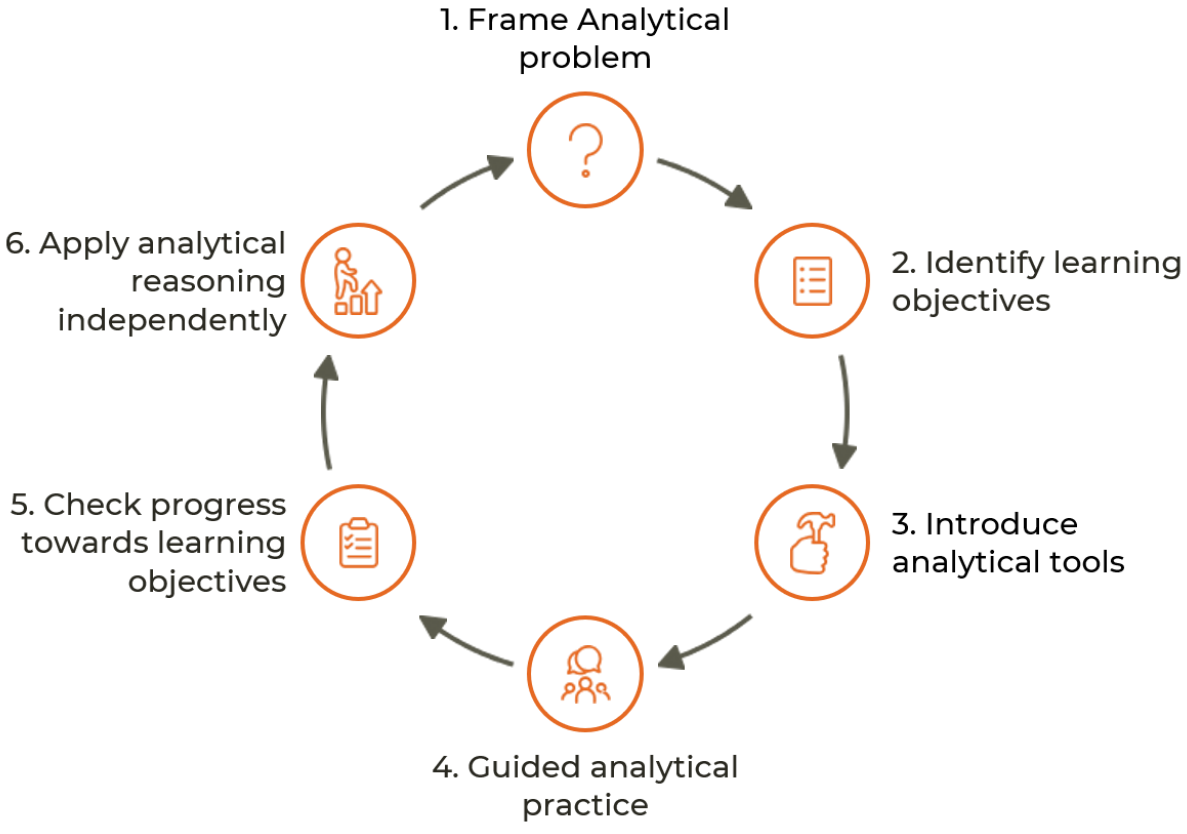
Teaching method

- The course utilizes problem-based-learning concept



- Students take **ownership** of their learning, identifying knowledge gaps, researching solutions, and evaluating their progress.
- Some problems are **open-ended**
- Effective **group work** is essential
- Tools include both data analysis methods and programming tools – use your resources

Problem based learning cycle



Evaluation methods

- The **bi-weekly assignments** are done **individually** and will contribute 50% to the course grade
  - You may **revise** your assignment based on the feedback within **one week** of receiving it. Your final grade will be based on the revised submission.

- The remaining 50% of the grade is defined by a **group project (3-4 students)** where you apply the learned techniques to your chosen engineering problem
  - Problem formulation and group formation on **October 7, 2026. Participation in the lecture is mandatory.**
  - Final report submission on **November 25, 2026**
  - Final presentation on **December 2, 2026**. **Participation in presentation is mandatory**.

- **PRESEMO Quizzes:** Not graded rather used to make sure the mastery of one topic before moving onto another (Participation in all quizzes save you 5% bonus)

- 1 point for **Course Feedback**

Evaluation methods

- If you are taking PNA, its final exam conflicts with this course's project demo. I suggest forming a project group with other students taking PNA so that I can arrange your group's demo on a separate day. Please see me after class.

- A discussion forum is available in the Mycourse page to facilitate formation of groups.

Workload (Approximation)

- Lectures: 24h (12x 2h/week)
- Studying materials and assignments: 50h (5 x 10h)
- Project work including report writing: 50h
- Final presentation preparation: 10 h

Using AI Tools

- AI allowed with attribution: You may use AI to complete tasks as needed. Any AI generated content **must be** cited.

- What matters most is **transparency**: if you use AI in solving an exercise, clearly state how and in which parts it was used. This helps us to understand your approach to learning.

- Keep in mind: AI is not a team member. You are responsible for the accuracy, quality, and integrity of your work. See full guidelines here.

Aalto AI tools

- Aalto AI assistant use cases
  - "Explain the given code step by step for someone new to Python."
  - "How can I modify this code to read the ship-speed and shaft-power data from a CSV file? Show what needs to change and explain how the CSV column names are used."

Python

```python
import pandas as pd
import matplotlib.pyplot as plt

# Example measurements from a ship
data = {
    "speed_knots": [10, 12, 14, 16, 18, 20],
    "shaft_power_kW": [1100, 1700, 2600, 3900, 5600, 7800]
}

# Convert the data into a table
ship_data = pd.DataFrame(data)

# Calculate the correlation
correlation = ship_data["speed_knots"].corr(
    ship_data["shaft_power_kW"]
)

# Visualize the relationship
plt.scatter(ship_data["speed_knots"],
            ship_data["shaft_power_kW"])

plt.xlabel("Ship speed (knots)")
plt.ylabel("Shaft power (kW)")
plt.title("Ship speed and shaft power")
plt.grid(True)
plt.show()

print(f"Correlation: {correlation:.2f}")
```

Aalto AI tools

- Aalto AI assistant use cases
  - "I have ship-speed data in knots and corresponding shaft-power data in kilowatts. How can I create a scatter plot in Python using Matplotlib?"
  - Label both axes with their units, add a title and grid, and explain each part of the code briefly.

You can use Matplotlib to create the scatter plot:

Python

```python
import matplotlib.pyplot as plt

# Example ship data
speed_knots = [10, 12, 14, 16, 18, 20]
shaft_power_kW = [1100, 1700, 2600, 3900, 5600, 7800]

# Create the scatter plot
plt.scatter(speed_knots, shaft_power_kW)

# Add labels and a title
plt.xlabel("Ship speed (knots)")
plt.ylabel("Shaft power (kW)")
plt.title("Relationship Between Ship Speed and Shaft Power")

# Add a grid and display the plot
plt.grid(True)
plt.show()
```

Here:

- `plt.scatter()` creates the scatter plot.
- `plt.xlabel()` and `plt.ylabel()` label the axes and show their units.
- `plt.title()` adds a descriptive title.
- `plt.grid(True)` makes the values easier to read.
- `plt.show()` displays the completed plot.

Each point represents one paired observation of ship speed and shaft power.

Keep your eye on the prize

Learn analytical decision-making under uncertainty and the ability to explain and justify it

Feedback

- In addition to the end-of-course feedback, you will have an opportunity to provide feedback each week.

- Your weekly feedback allows us to make immediate adjustments and has proved very useful in creating an effective and inclusive learning environment.

- Some feedback from previous years
  - *"I like – the combination of theory and coding tasks in the class. The pace, the content, the coding is fun."*
  - *"I like – the way it was designed, starts from basic and touches advance concepts as well."*
  - *"I wish – clearer assignment instructions."*

Resources

- Lecture slides and Jupyter notebooks used in the class will be the primary resource

